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AWaveFormer: Audio Wavelet Transformer Network for Generalized Audio Deepfake Detection

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Abstract

Generative neural vocoders like HiFi-GAN and WaveNet generate synthetic speech with high perceptual realism. AWaveFormer presents a unified generalized deepfake detection network that decomposes raw audio into time-frequency subbands via Stationary Wavelet Transform (SWT) and models long-range temporal artifacts using a multi-head Wavelet Transformer encoder.

Key Methodology & Architecture

Stationary Wavelet Transform (SWT), Multi-Head Self-Attention Transformer, Cross-Scale Feature Fusion, Binary Cross-Entropy classification.

Datasets & Benchmark Environments

ASVspoof 2019 LA/DF, WaveFake, In-the-Wild Deepfake Audio Dataset.

Performance Metrics & Graph Axis Parameters

Reported Results: Equal Error Rate (EER) = 0.82%, Tandem Detection Cost Function (t-DCF) = 0.024, Accuracy = 99.18%.

Graph Parameter Mapping: X-axis: False Alarm Rate (FAR in %, 0% to 10%); Y-axis: Miss Detection Rate (MDR in %, 0% to 10%) [ROC Curve].

Comparative Analysis

Advantages (Pros)	Limitations (Cons)
Exceptional cross-dataset generalization across unseen text-to-speech (TTS) engines.	Wavelet decomposition adds non-negligible memory footprint.

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